MILITARY SPECIFICATION SHEET

# Gases, Calibrating, and Mixtures; 80% Nitrogen, 10% Helium, 10% Oxygen

This specification is approved for use by San Antonio ALC/SFRM, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the gas mixture described herein shall consist of this document and the latest issue of specification MIL-G-83794.

Grade Designation: C

| Component | Requirement (% by Volume) | Preparation Tolerance (% by Volume) | Test Method (Paragraph) |
|---|---|---|---|
| Helium | 10.0 | ±0.5 | 4.6.5 |
| Oxygen | 10.0 | ±0.5 | 4.6.4 |
| Nitrogen | 80.0 | ±1.0 | 4.6.5 |

| Contaminant | Requirement (ppm by Volume, Maximum) | Test Method (Paragraph) |
|---|---|---|
| Water | 10 | 4.6.1 |
| Total Hydrocarbons (THC) as Methane | 10 | 4.6.2 |
| Total Impurities except argon, includes $H_2$, Ne, CO, $CO_2$, $H_2O$, THC, Kr, Xe, and $N_2O$ | 50 | 4.6.1, 4.6.2, 4.6.5, 4.6.6, 4.6.7, and 4.6.8 |

Lot: A lot shall consist of one of the following:

Each set of cylinders filled on the same manifold at the same time.

The gas mixture which is held in a single storage tank and subsequently withdrawn to fill cylinders. The product shall be homogeneous at the time of withdrawal and shall not be added to while being withdrawn. After each addition to the storage tank, the contents shall constitute a separate lot.

Product Conformance Tests: Samples shall be drawn at random from each lot at the point of filling and tested for components and contaminants prior to shipment. The number of samples drawn shall be as follows:

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| Number of Containers Per Lot | Number of Containers To Be Sampled |
|---|---|
| 1-8 | 1 |
| 9-40 | 2 |
| 41-70 | 3 |
| Over 70 | 4 |

Cylinder: DOT 3A or 3AA

Valve: MIL-V-2/6, except the valve body shall not be marked.

Title and Color Code for Government Owned Cylinders:

TITLE

80% Nitrogen, 10% Helium, 10% Oxygen

| LOCATION ON CYLINDER | | | |
|---|---|---|---|
| Top A | Band B | Band C | Body |
| Gray | Green | Gray | Gray |

The above data supplements the requirements of MIL-STD-101.

Intended Use: This gas mixture is used to provide a suitable operating atmosphere for an electronic system.

CUSTODIAN:
Air Force - 68

PREPARING ACTIVITY:
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